

CHAPTER 6

PICTORIAL DRAWING

6.1 *Introduction*

Pictorial view is a three dimensional one which is used to visualize an object in one view. On the other hand, the orthographic views such as, top view, front view, side view etc. which are separated from each other provide necessary information about dimensions, material, surface finish etc. to manufacture the object. With the help of pictorial view complicated engineering drawings can easily be communicated to the people who do not have sufficient training in understanding the orthographic views. A pictorial view provides the main dimensions of the object only. It cannot be used as a working drawing. It is used only to visualize the object. Pictorial drawing is classified as:

- (1) Axonometric
- (2) Oblique
- (3) Perspective

6.2 *Perspective View*

The perspective drawing is relatively complicated to produce and is used mainly by the architects. A perspective projection is not suitable for working drawing. An example of a perspective drawing has been shown in Figure 6.1. Perspective drawing is the view drawn looking an object with normal vision. When an object goes away from the observer, it appears smaller and ultimately vanishes at a point. However, the engineers use only the oblique and axonometric drawing.

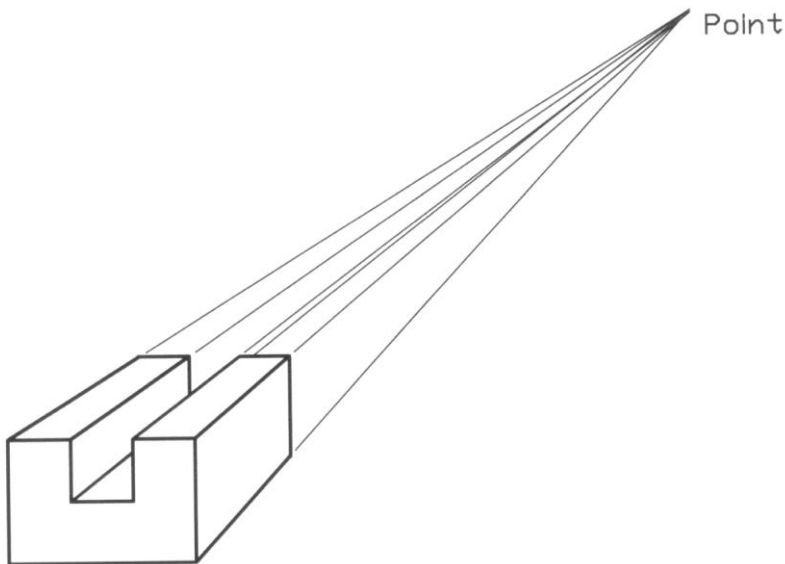


Figure 6.1: One Point Perspective View

6.3 Axonometric Projection

In the axonometric projection the three faces of an object are seen on the plane of projection (picture plane) and the projection lines from the object are perpendicular to the plane of projection. To understand an axonometric projection, a cube is considered. If the cube is placed behind a vertical transparent plane in such a way that one of the faces of the cube is parallel to the plane, the view on the plane based on orthographic projection will be of square shape that is represented by the shaded view as shown in Figure 6.2. Top and right side views have also been shown in the figure.

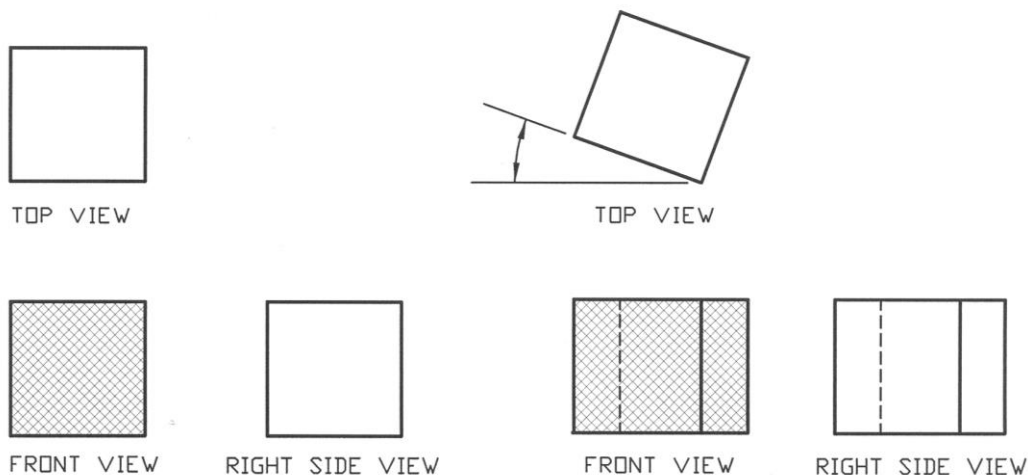


Figure 6.2: Views with One Face Parallel To the Projection Plane

Figure 6.3: Views after Rotation About Vertical Axis

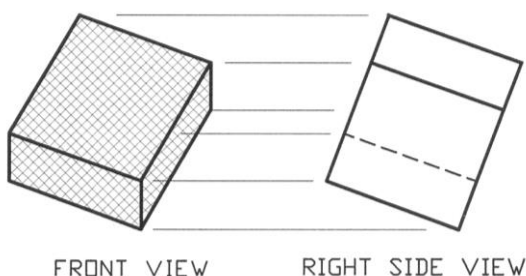


Figure 6.4: Views after Rotation about Horizontal Axis (Tilting)

If the cube is now rotated by any angle less than 90° about the vertical axis, the view on the plane will be similar to that represented by the shaded view as shown in Figure 6.3. Top and side views have also been shown. Now if the cube just from this position is rotated by any angle less than 90° about the horizontal axis that is parallel to the projection plane, the view on the plane will appear similar to that shown by the shaded view in Figure 6.4. Right side view has also been presented in

this figure. The angle at which the cube is rotated may be of infinite number; as a result the number of axonometric positions may be infinite. It is usual practice to use three types of positions. Accordingly axonometric projections are of three types:

- (1) Isometric
- (2) Dimetric
- (3) Trimetric

6.4 Isometric Projection

To obtain the isometric projection, the cube is rotated by 45° about the vertical axis (Figure 6.5) from the position as shown in Figure 6.2, and then rotated about the horizontal axis, which is parallel to the projection plane in such a way that the top face of it makes a slope of $35^\circ 16'$ approximately (Figure 6.6). In this position the diagonal OE passing through O, is perpendicular to the projection plane and the three mutually perpendicular edges OA, OB and OC make angles of 120° with each other. The projection of this particular position as represented by the front view in Figure 6.6 is called the isometric projection. (iso means same and metric means measurement, thus isometric means same measurement).

According to the geometrical principle when the top surface of the cube makes a slope of $35^\circ 16'$, then the angles between the edges OA, OB and OC of the cube become 120° on the projection plane. The length OA as represented by the right side view of Figure 6.6, is the actual length (100%) and its projected length OA as shown in the front view in the same figure is shortened to 81% approximately. The isometric axes OA, OB and OC are foreshortened equally because they are at the same angle of 120° to the projection plane.

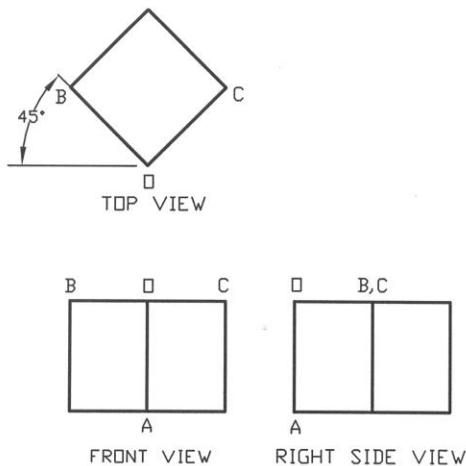


Figure 6.5: Views after Rotation About Vertical Axis by 45°

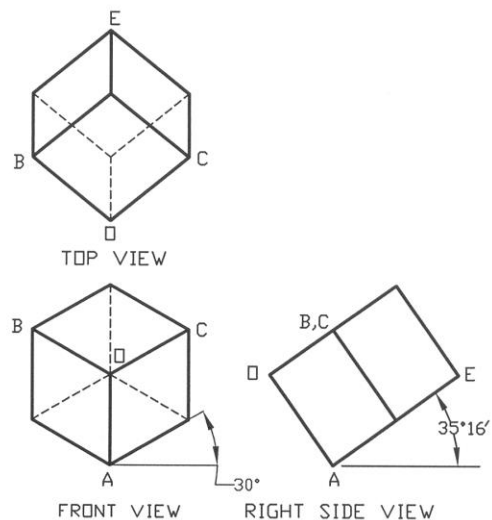


Figure 6.6: Views after Rotation About Horizontal Axis by $35^\circ 16'$

In Figures 6.7 and 6.8 an isometric projection and isometric drawing are shown. An isometric drawing differs from an isometric projection in that the isometric drawing is done in actual size while the isometric projection is drawn foreshortened. If the actual size is 100%, the size in the

isometric projection is $100\cos(35^\circ 16')$ i.e. 81% approximately. As a matter of fact it is usually convenient to make the isometric drawing rather than the isometric projection.

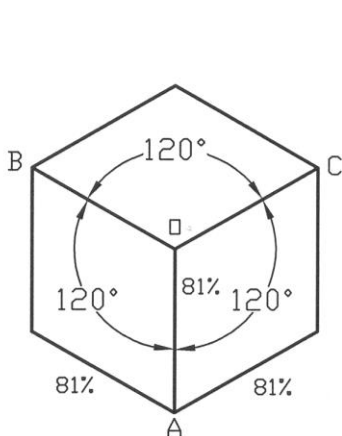


Figure 6.7: Isometric Projection

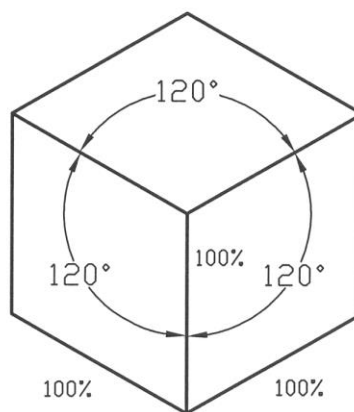


Figure 6.8: Isometric Drawing

Here the lines OA, OB and OC are called isometric axes. Any line parallel to the isometric axis is called isometric line. The line which, is not parallel to the isometric axis, is called non-isometric line. The planes of the faces of the isometric cube and all other planes, which are parallel to them, are called isometric planes.

In the isometric projection, the isometric lines can be drawn using isometric scale (Figure 6.9). The isometric line is shortened to 81% approximately, which is obtained from the isometric scale directly.

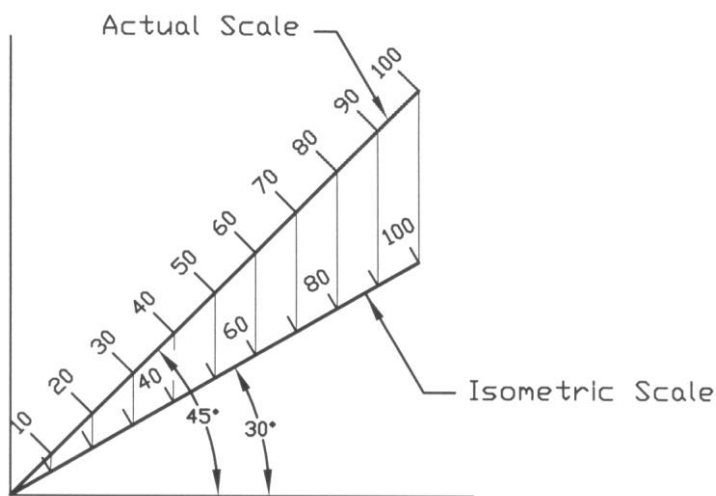


Figure 6.9: Isometric Scale

In dimetric projection, the two angles (θ_1 , θ_2) as shown in Figure 6.10 are kept equal and may be any angle between 0° and 45° except 30° . While in the trimetric projection the angles (θ_1 , θ_2) are not equal; sum of them is less than 90° but neither angle is 0° . The comparisons of the isometric, dimetric and trimetric projections have been shown in Figure 6.10.

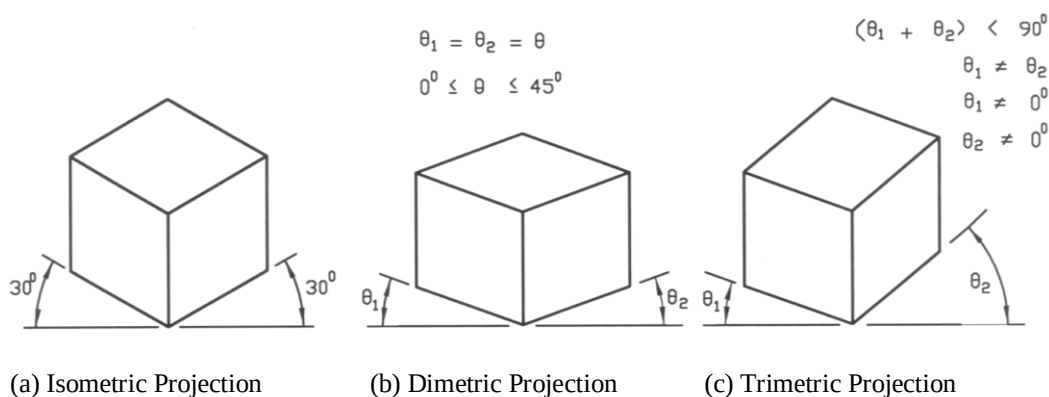


Figure 6.10: Types of Axonometric Projection

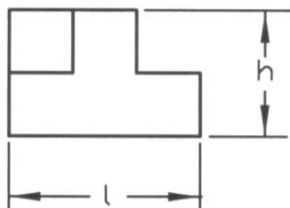
6.5 Making Isometric View

A simplified procedure is provided below to draw an isometric view from the orthographic projection. In Figure 6.11, the orthographic projection of an object is given from where an isometric view will be drawn. The steps as mentioned below are to be followed in order to make the isometric view.

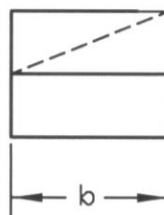
- (1) An isometric parallelepiped is drawn (Figure 6.12) in such a way that the length (l), breadth (b) and height (h) of the parallelepiped are respectively equal to those of the orthographic projection (Figure 6.11). The lines will be very thin so that they can be erased easily when required.
- (2) Now the views are drawn on the respective side of the isometric parallelepiped keeping appropriate relationship with each other (Figure 6.13). In drawing views the unnecessary lines may be avoided otherwise, the drawing will be clumsy. The horizontal lines of the view will be parallel to the isometric axes OB and OC while the vertical lines will be parallel to the isometric axis OA . Each line will be thin.
- (3) Next shifting of the necessary surfaces is made keeping conformity with the views. Surface $a-b-c-d$ has been shifted to the new position $a'-b'-c'-d'$ and surface $p-q-r-s$ has been shifted to the new position $p'-q'-r'-s'$ (Figure 6.14).
- (4) The unnecessary lines are now erased and the isometric view is completed making the lines thick (Figure 6.15).
- (5) Finally from the generated isometric view the orthographic projection may be verified to confirm the drawing whether it has been done properly.



TOP VIEW



FRONT VIEW



RIGHT SIDE VIEW

Figure 6.11: Orthogonal Projection

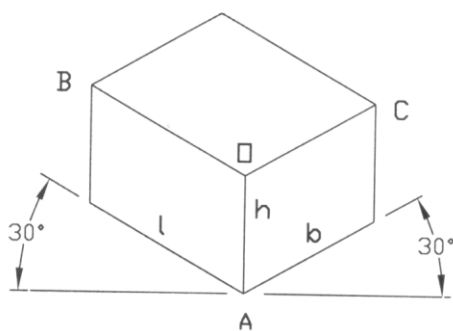


Figure 6.12: Isometric Parallellepiped

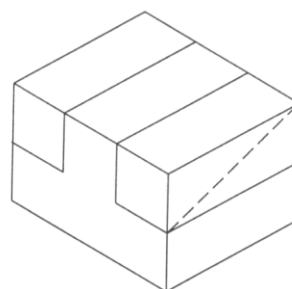


Figure 6.13: Views on Each Side

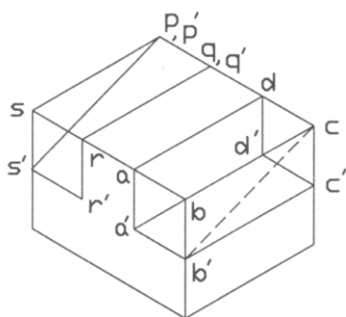


Figure 6.14: Shifting of Surfaces

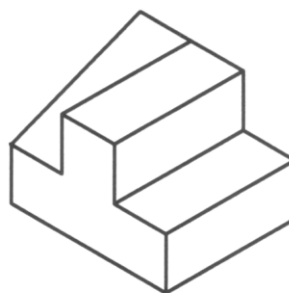
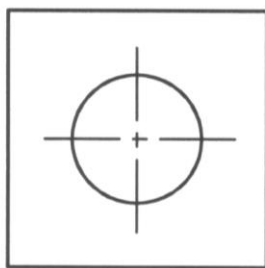


Figure 6.15: Isometric View

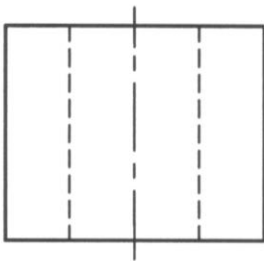
6.6 Making Isometric View with Circular Features

A circle in the orthographic projection is transformed into the shape of an ellipse. In Figure 6.16 the orthographic projection of an object with circular feature is provided. To make the isometric view the steps as mentioned below are followed.

- (1) A square touching the circle is drawn with sides equal to the diameter of the circle on the orthographic projection (Figure 6.17).
- (2) Now the isometric parallelepiped (Figure 6.18) is drawn as done earlier.
- (3) Next the views on the surfaces of the parallelepiped (Figure 6.19) are drawn with the square of the circle omitting the circle itself. The square is turned into the shape of a rhombus. The unnecessary lines may be avoided.
- (4) Now the lines on the rhombus are drawn as shown in Figure 6.20. Then the four centers c_1 , c_2 , c_3 , and c_4 are located.
- (5) The two arcs are drawn with radii c_1b and c_2a with respect to the centers c_1 and c_2 respectively. Next two other arcs are drawn with radii c_3a and c_4b with respect to the centers c_3 and c_4 respectively (Figure 6.21).
- (6) Now the unnecessary lines are erased and the isometric view is completed making the lines thick (Figure 6.22).



TOP VIEW



FRONT VIEW

Figure 6.16: Orthographic Projection

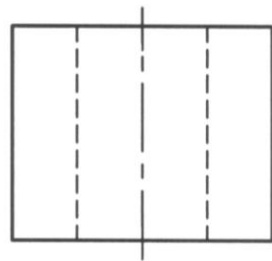
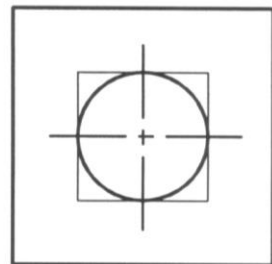


Figure 6.17: Square Touching Circle

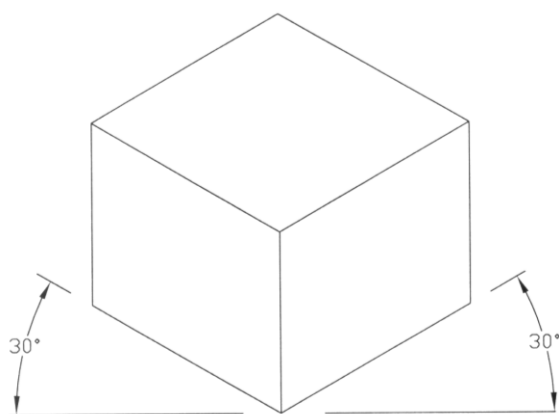


Figure 6.18: Isometric Parallelepiped

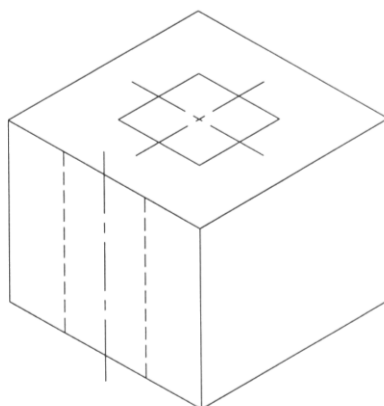


Figure 6.19: Views on Faces of Parallelepiped

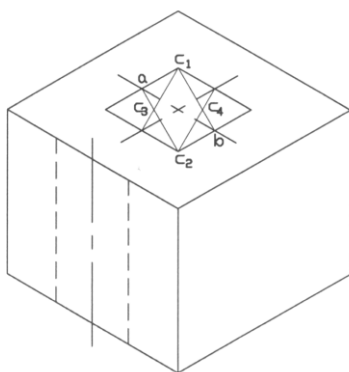


Figure 6.20: Location of Centers

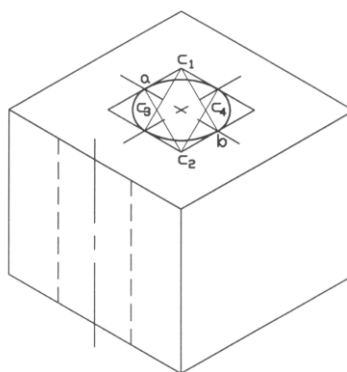


Figure 6.21: Drawing Arcs

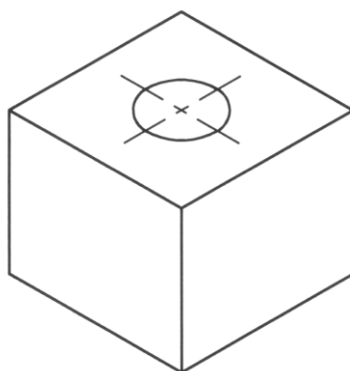


Figure 6.22: Isometric Circle

The orthographic projection of an object incorporating the circular feature is shown in Figure 6.23. The various steps are represented graphically in Figures 6.24 to 6.31 in order to generate an isometric view. This is provided as a further example.

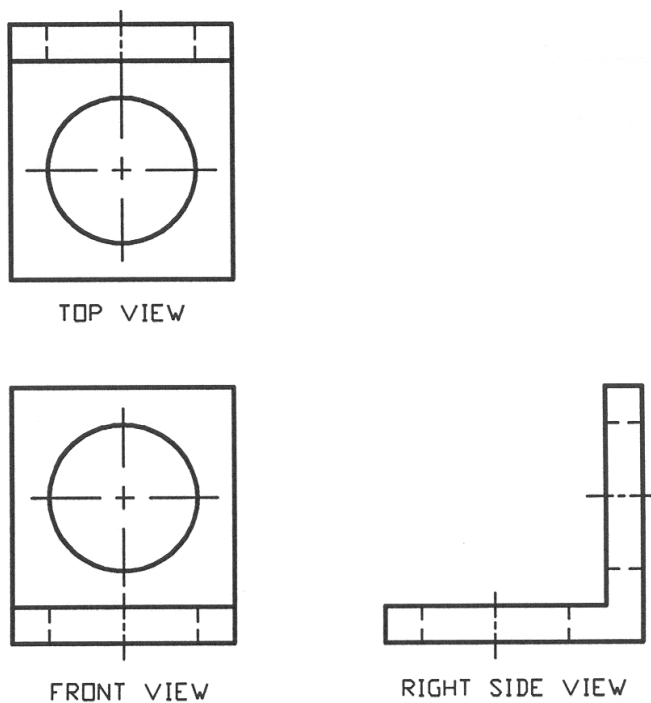


Figure 6.23: Orthographic Projection

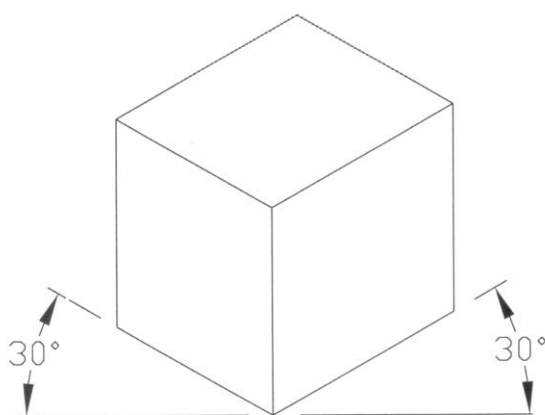


Figure 6.24: Making Parallelepiped

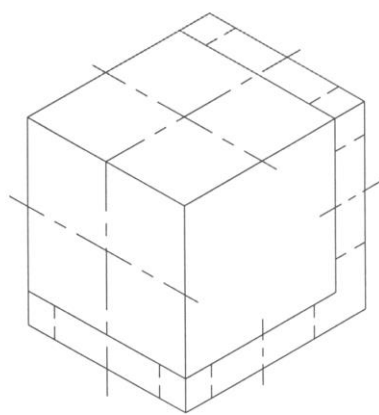


Figure 6.25: Views on Faces

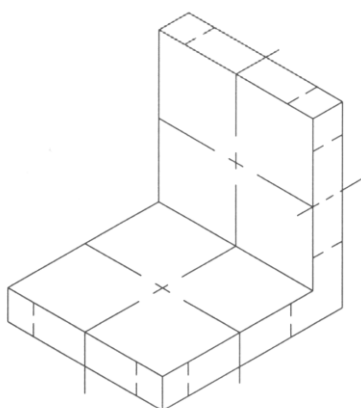


Figure 6.26: Shifting Faces

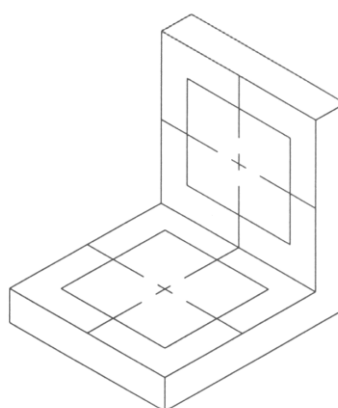


Figure 6.27: Rhombuses for Circles

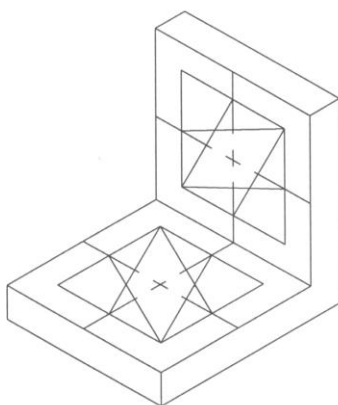


Figure 6.28: Locating Centers

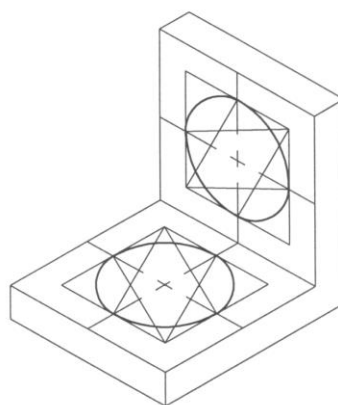


Figure 6.29: Making Ellipses

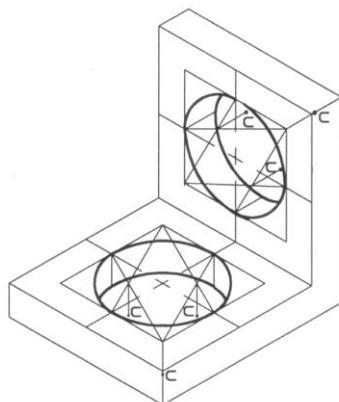


Figure 6.30: Ellipses on Other Faces

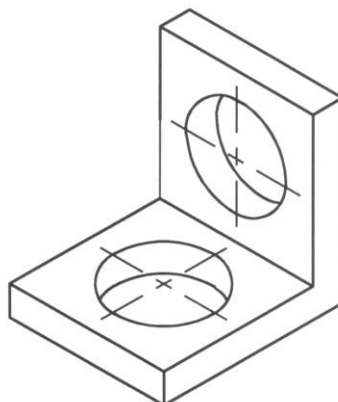


Figure 6.31: Complete Isometric View

6.7 Oblique Projection

In generating an isometric projection, one looks in a perpendicular direction through the projection plane at an object, which is positioned in rotating condition with respect to the projection plane (rotated about vertical axis and tilted). But in producing oblique projection opposite phenomenon occurs. The object is positioned with its major face parallel to the projection plane. Now an observer looks at the object through the projection plane at an oblique angle unlike in the case of isometric projection. In Figure 6.32 an oblique projection has been presented.

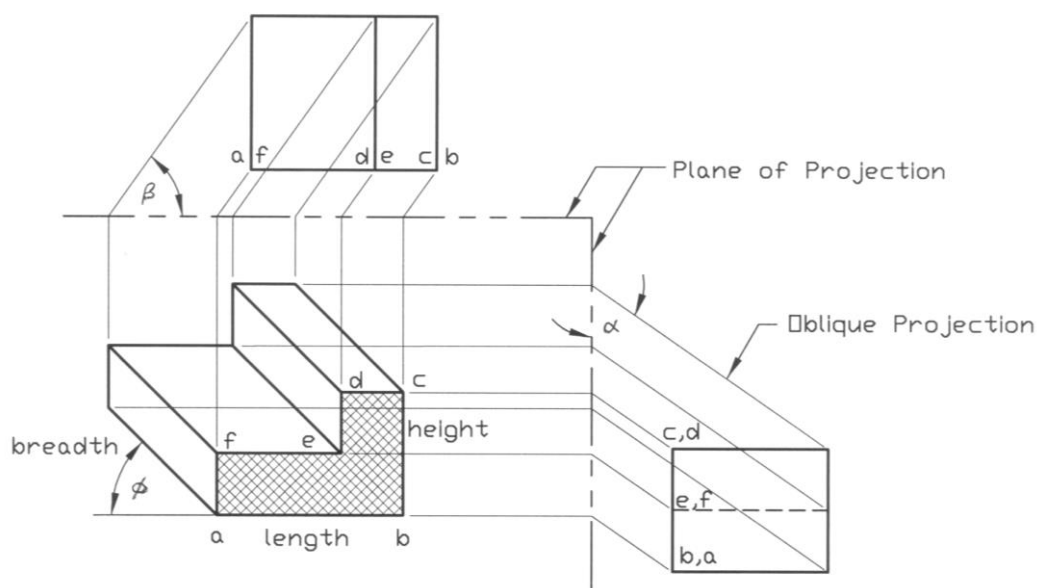


Figure 6.32: Oblique Projection

The axes of length and height are perpendicular to each other as shown in the figure. The axis of breadth can vary in angle ϕ and in length. Angle ϕ and the breadth of the object may be any value but usual practice is to keep ϕ at 30° or 45° for convenience. The surface of the object, which is parallel to the frontal plane of projection, represents the true shape as shown by the shaded area in the figure. This is the basic character of the oblique projection. As a result any surface of the object containing circle, curve or irregular shape, which is parallel to the frontal plane represents the exact shape. That is why the oblique projection is used mainly for the objects containing circle, curve or irregular features on one face or on parallel faces. For objects of this type oblique projection is easier to draw and give dimension.

In Figure 6.33 some oblique views are provided. When the breadth equals the full size (L), it is called cavalier and when the breadth equals the half size ($.5L$), it is called cabinet. Cabinet drawing looks pleasant compared to cavalier one. In case of the breadth in between the half and the full size ($.5L < l < L$), it is known as general.

In making an oblique view three mutually perpendicular axes passing through O (Figure 6.35) are chosen, one is horizontal, one is vertical and the other one is at an angle. On the three axes the height (h), width (w) and depth (d) are chosen in accordance with the orthographic projection

(Figure 6.34). Now the centers are located and the view is completed. The interesting point can be noted that, as the frontal face contains the circle, no transformation into ellipse is necessary. All the circles in this feature remain as they are. It makes the drawing easier.

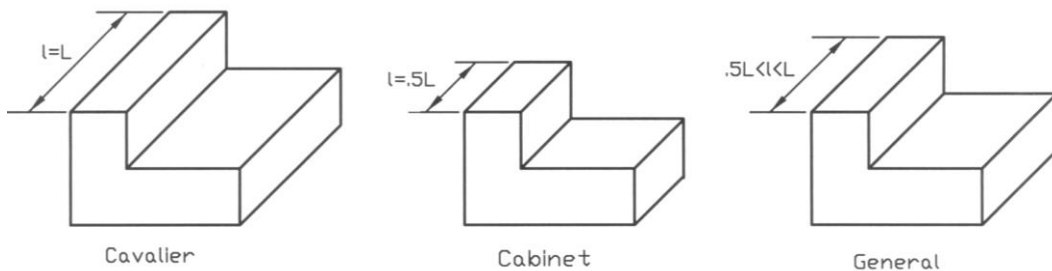


Figure 6.33: Oblique Views

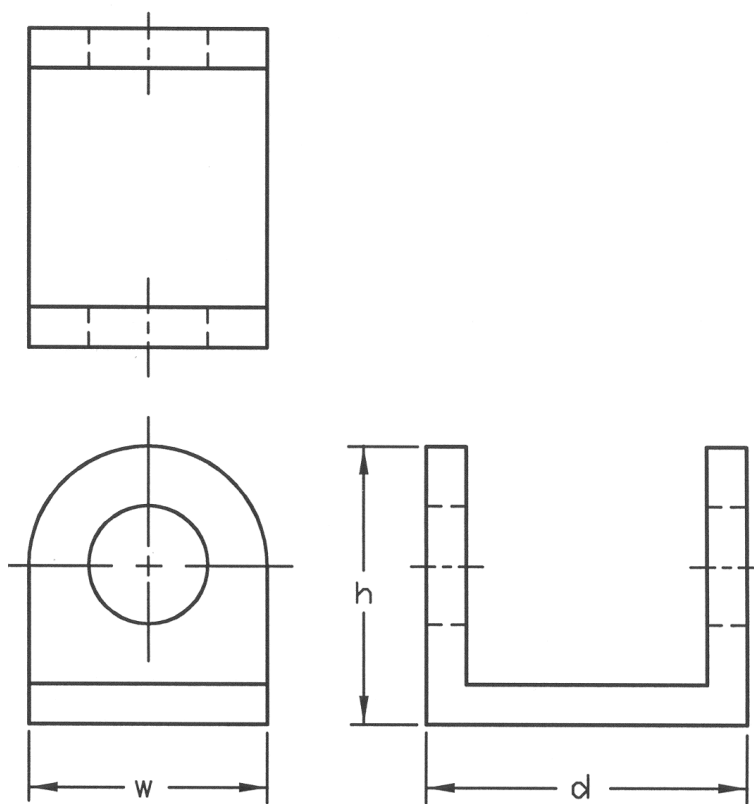


Figure 6.34: Orthographic Projection

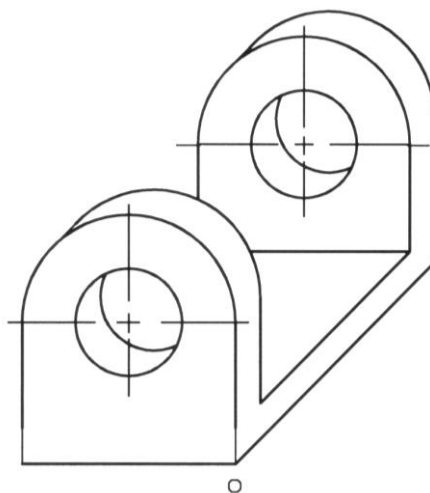


Figure 6.35: Oblique View with circular Feature

Example Problems

Note: For solutions see the following section of *Solutions for Example Problems*.

Prob. 6.1- 6.24: Draw the isometric sketches of the objects from the following views as shown in Fig. P6.1 to P6.24. The free hand Sketches may be done. Here the scaled drawings are provided.

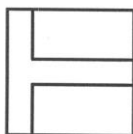


Fig. P6.1

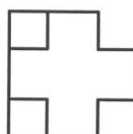


Fig. P6.2

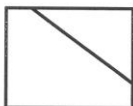
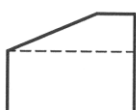
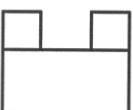


Fig. P6.3



Fig. P6.4



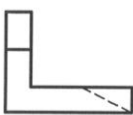
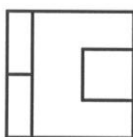


Fig. P6.5

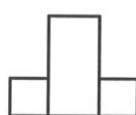
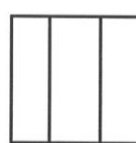


Fig. P6.6

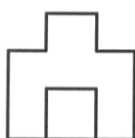
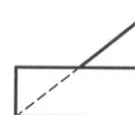


Fig. P6.7

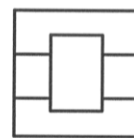
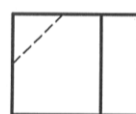


Fig. P6.8

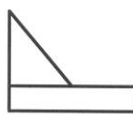


Fig. P6.9

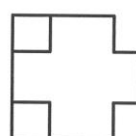
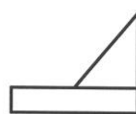


Fig. P6.10

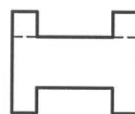
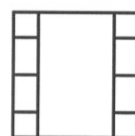


Fig. P6.11

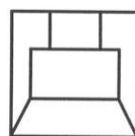


Fig. P6.12

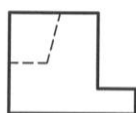
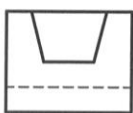


Fig. P6.13

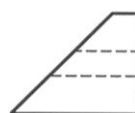
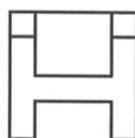


Fig. P6.14



Fig. P6.15

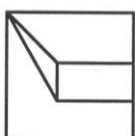


Fig. P6.16

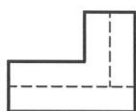
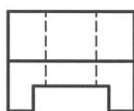
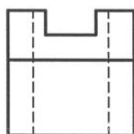


Fig. P6.17

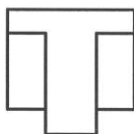


Fig. P6.18

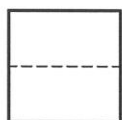


Fig. P6.19

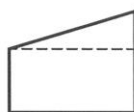


Fig. P6.20

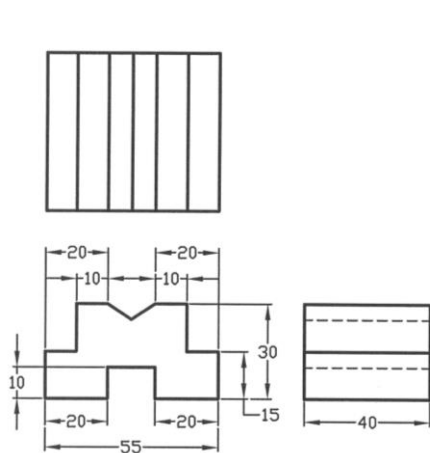


Fig. P6.27

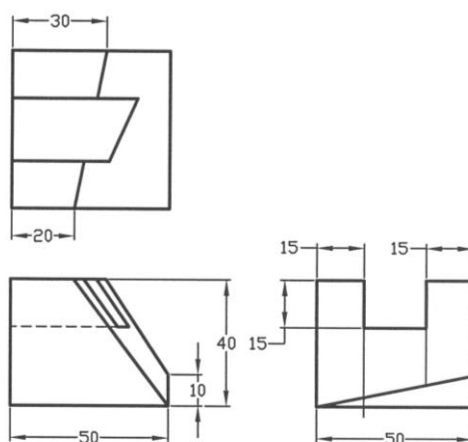


Fig. P6.28

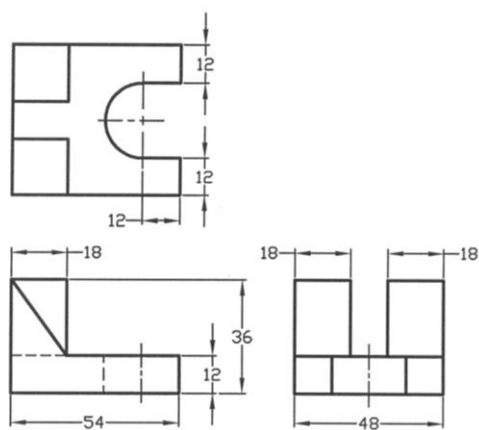


Fig. P6.29

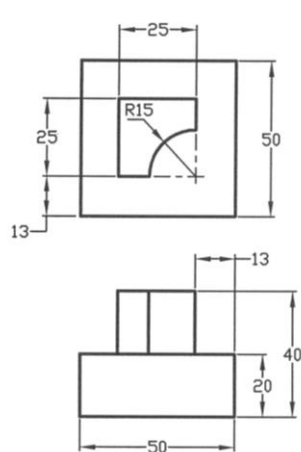


Fig. P6.30

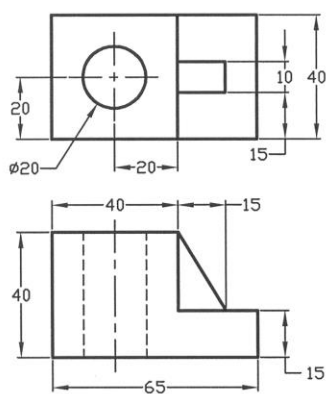


Fig. P6.31

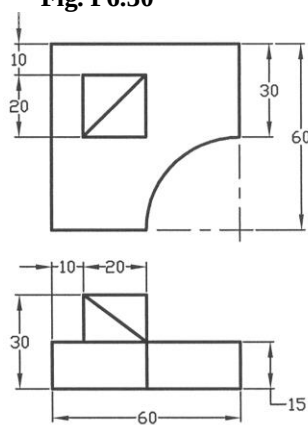


Fig. P6.32

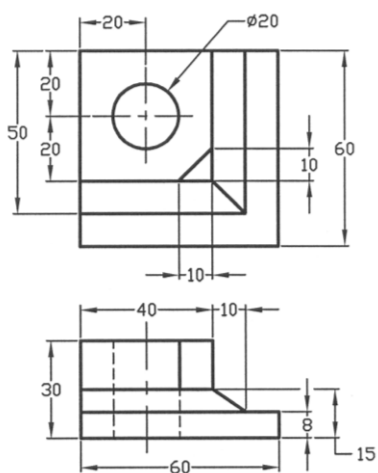


Fig. P6.33

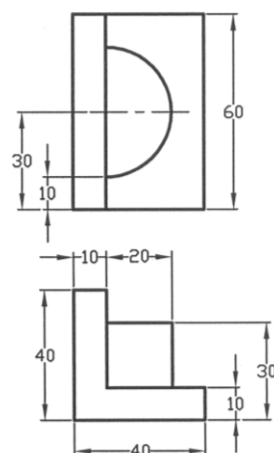


Fig. P6.34

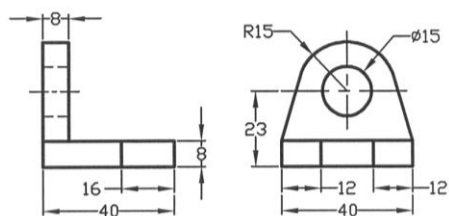


Fig. P6.35

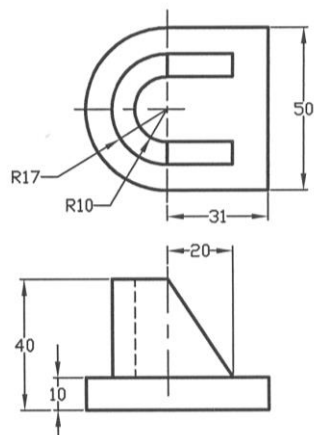


Fig. P6.36

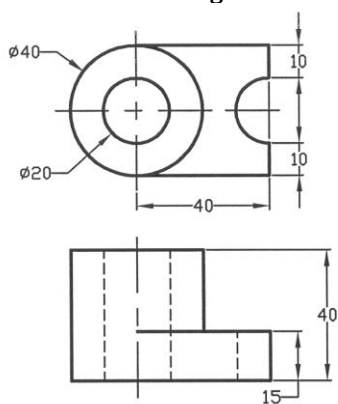


Fig. P6.37

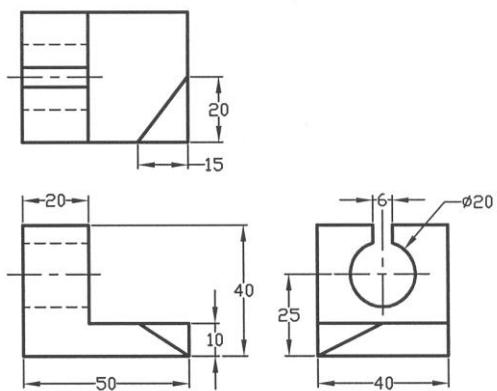


Fig. P6.38

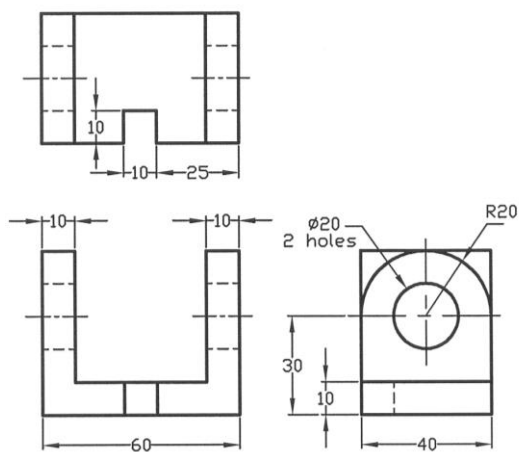


Fig. P6.39

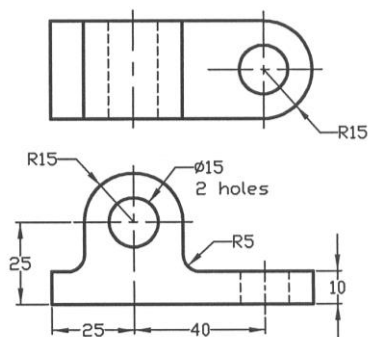


Fig. P6.40

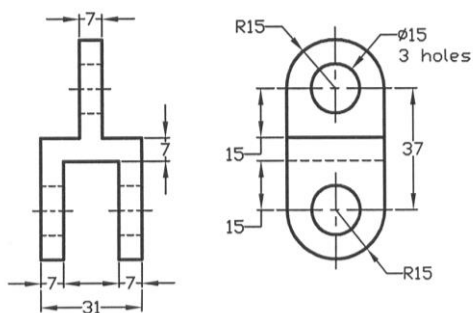


Fig. P6.41

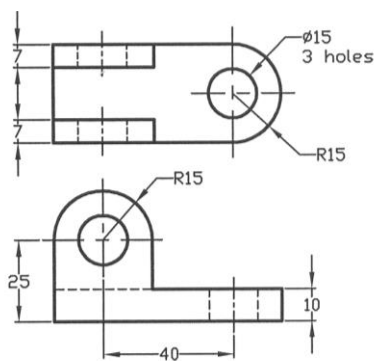


Fig. P6.42

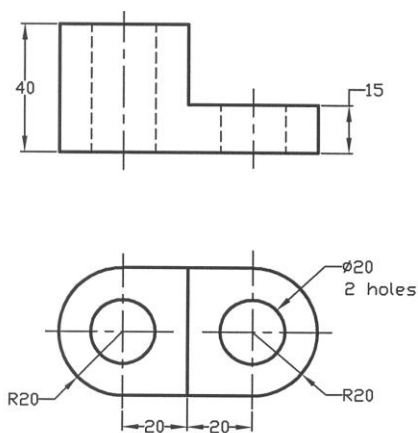


Fig. P6.43

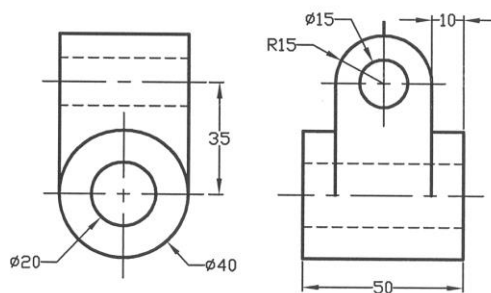


Fig. P6.44

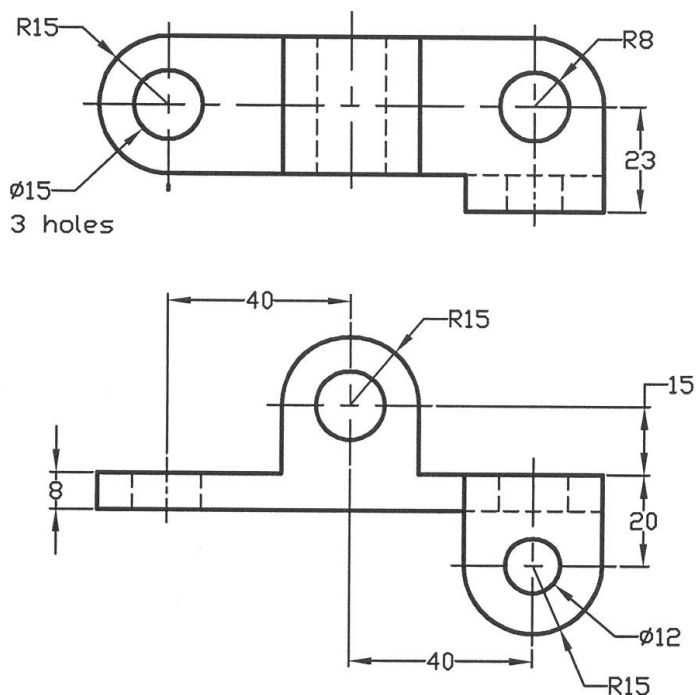


Fig. P6.45

Prob. 6.46- 6.51: Draw the oblique views (cavalier) of the objects from the following views as shown in Figures P6.46 to P6.51.

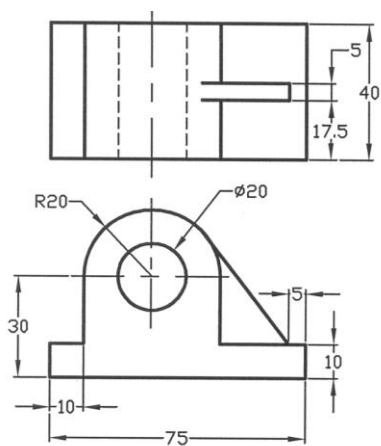


Fig. P6.46

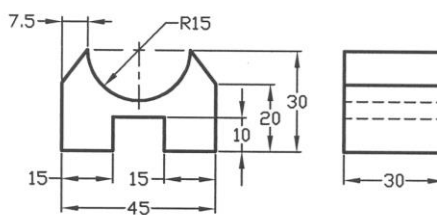


Fig. P6.47

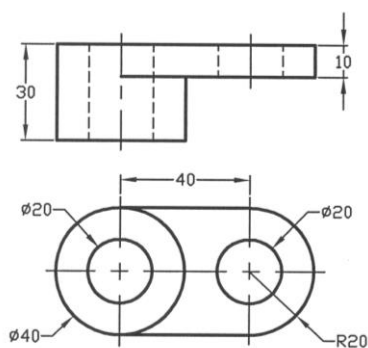


Fig. P6.48

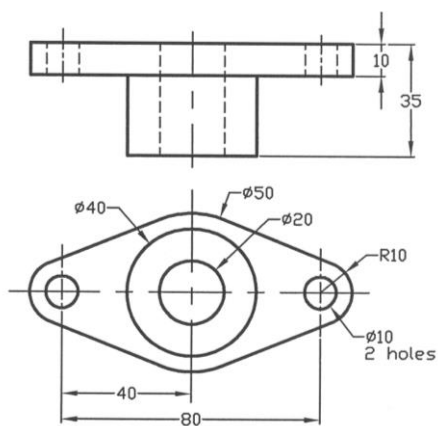


Fig. P6.49

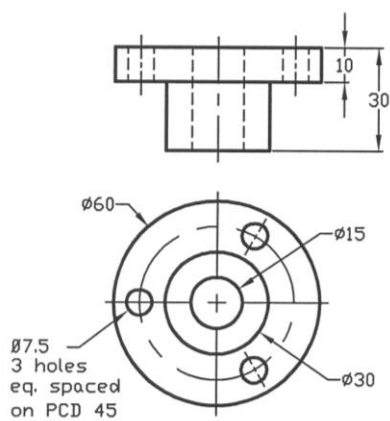


Fig. P6.50

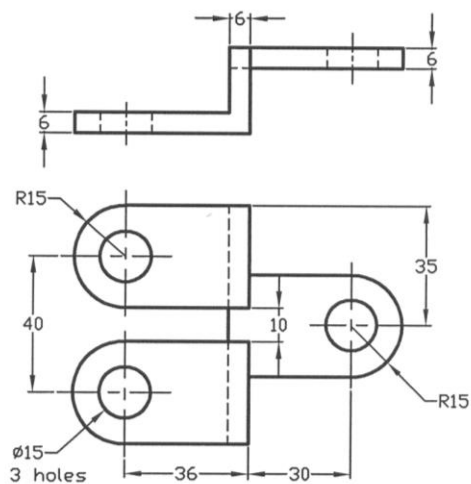
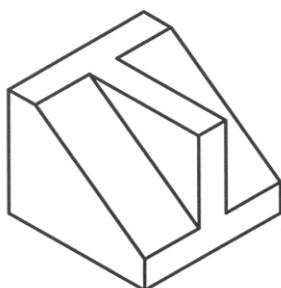
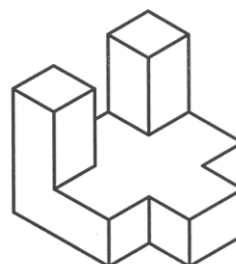


Fig. P6.51

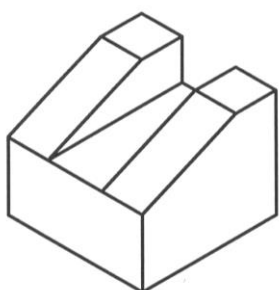
Solutions for Example Problems



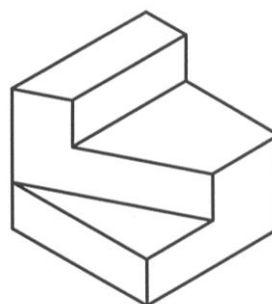
Solution of P6.1



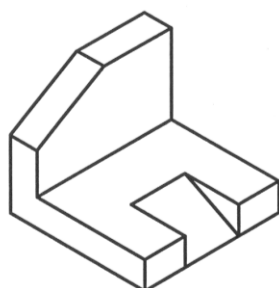
Solution of P6.2



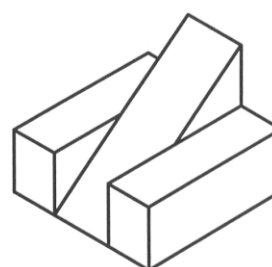
Solution of P6.3



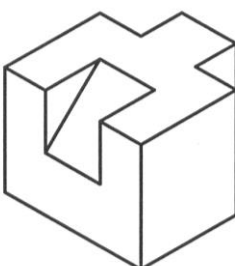
Solution of P6.4



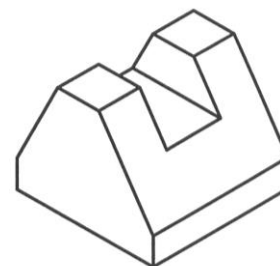
Solution of P6.5



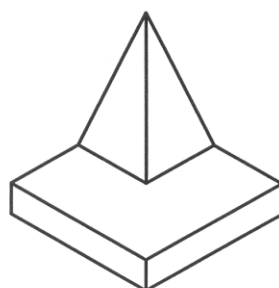
Solution of P6.6



Solution of P6.7



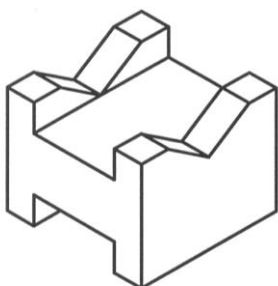
Solution of P6.8



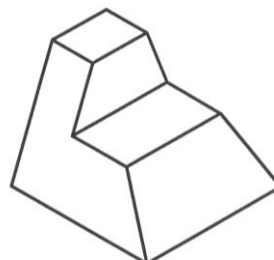
Solution of P6.9



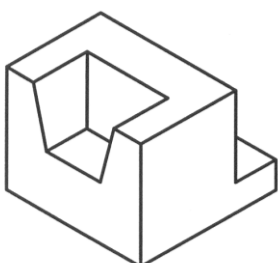
Solution of P6.10



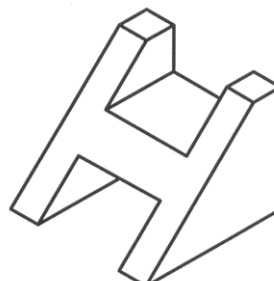
Solution of P6.11



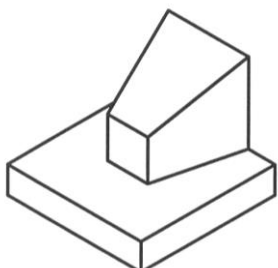
Solution of P6.12



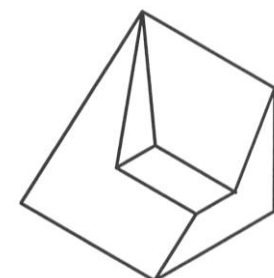
Solution of P6.13



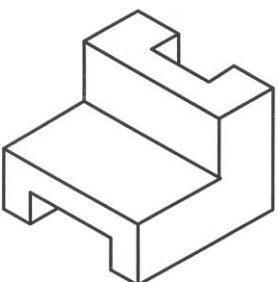
Solution of P6.14



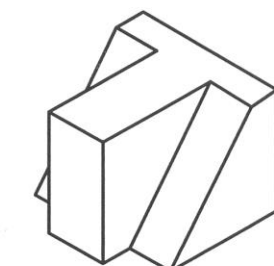
Solution of P6.15



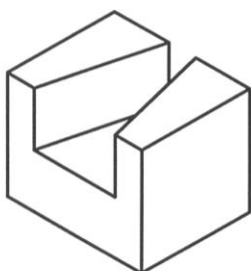
Solution of P6.16



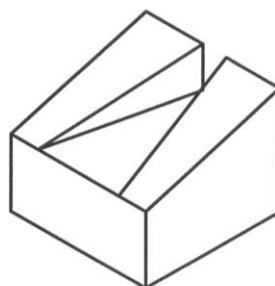
Solution of P6.17



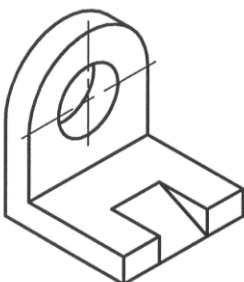
Solution of P6.18



Solution of P6.19



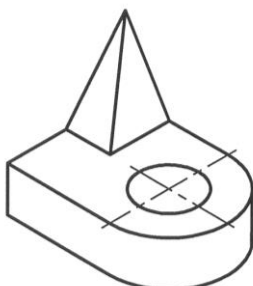
Solution of P6.20



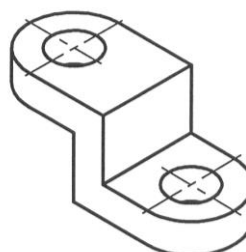
Solution of P6.21



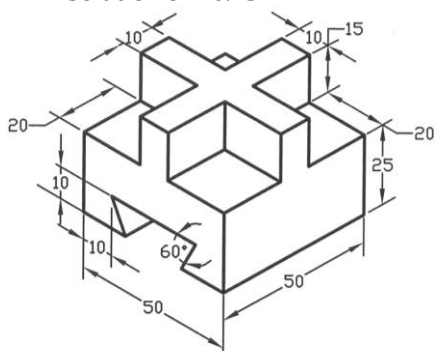
Solution of P6.22



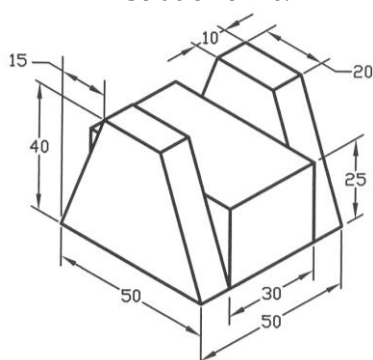
Solution of P6.23



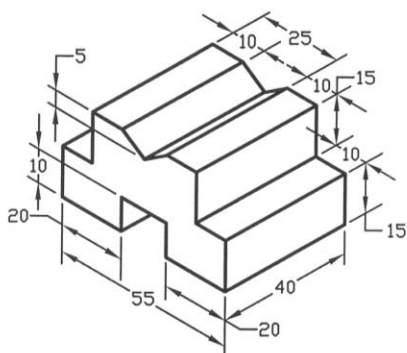
Solution of P6.24



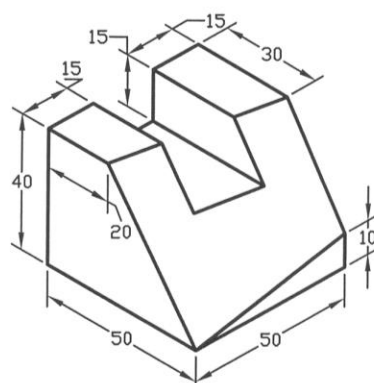
Solution of P6.25



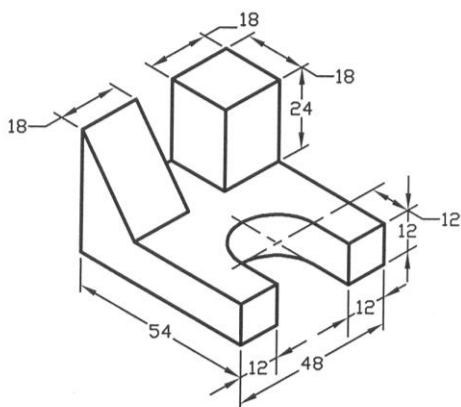
Solution of P6.26



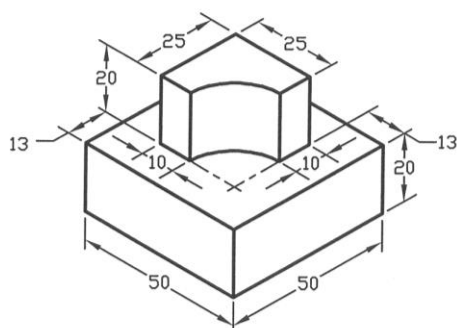
Solution of P6.27



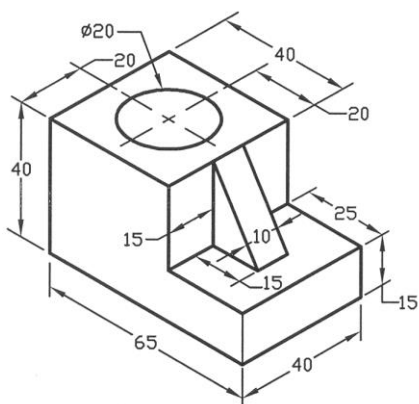
Solution of P6.28



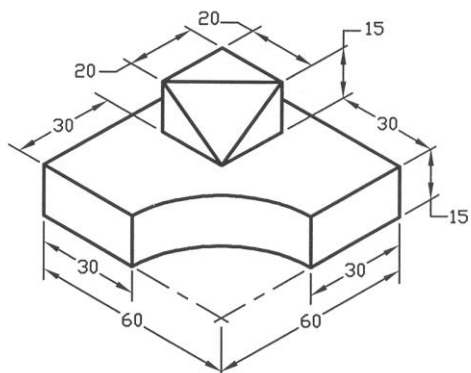
Solution of P6.29



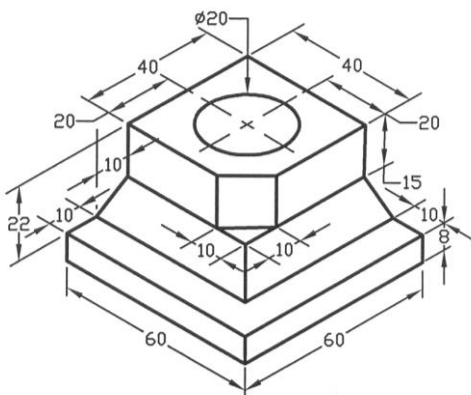
Solution of P6.30



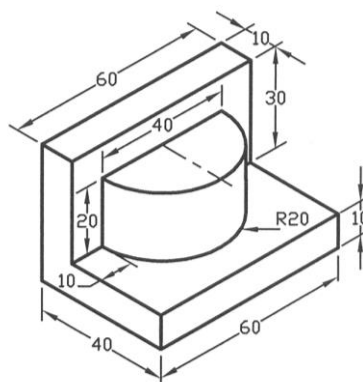
Solution of P6.31



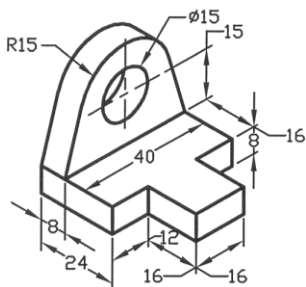
Solution of P6.32



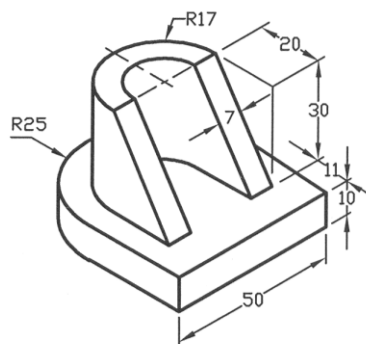
Solution of P6.33



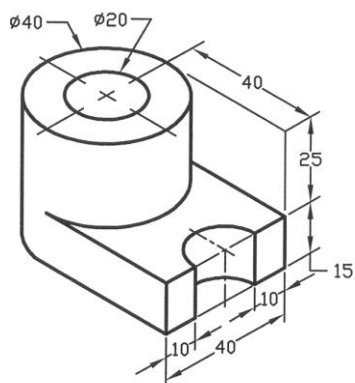
Solution of P6.34



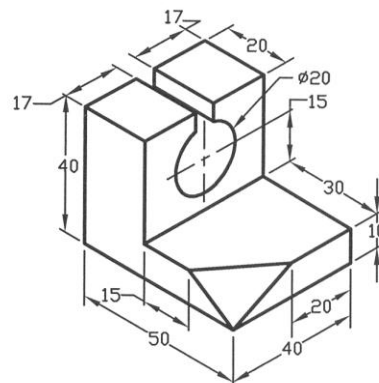
Solution of P6.35



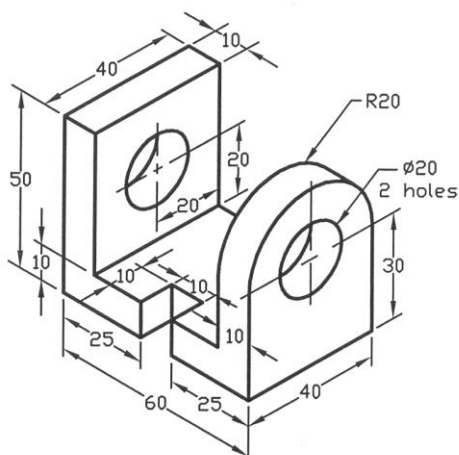
Solution of P6.36



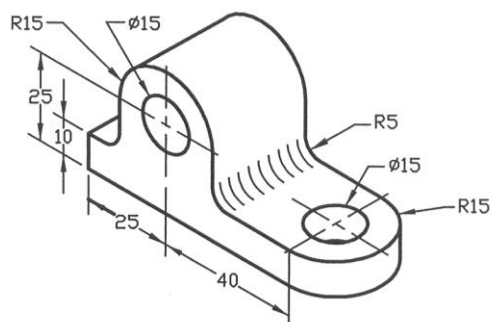
Solution of P6.37



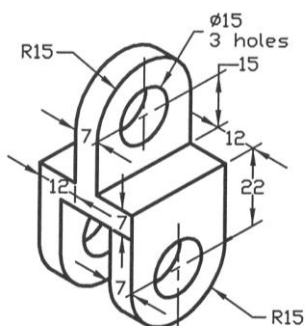
Solution of P6.38



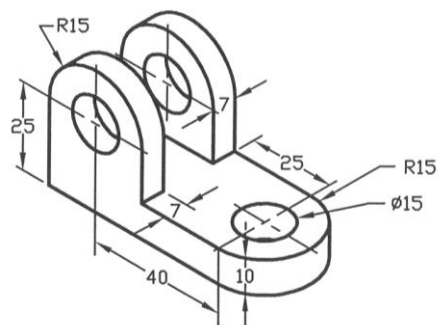
Solution of P6.39



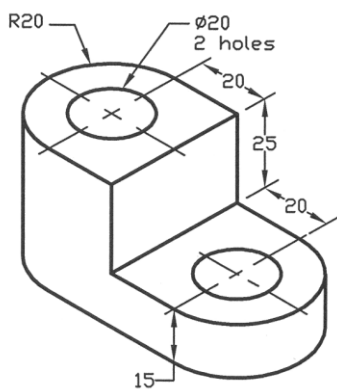
Solution of P6.40



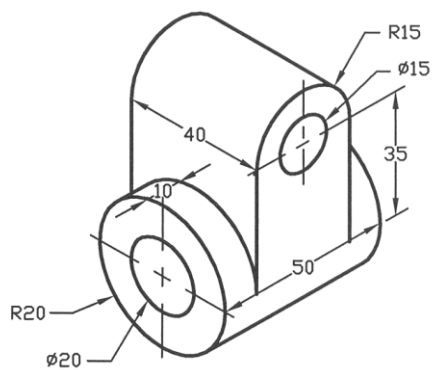
Solution of P6.41



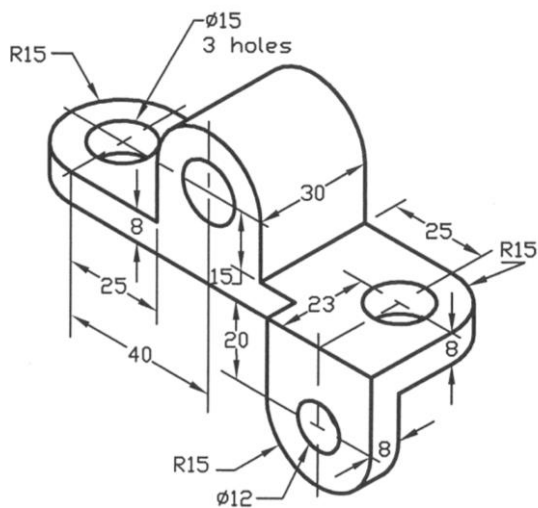
Solution of P6.42



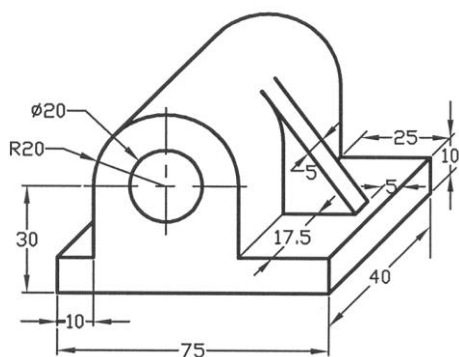
Solution of P6.43



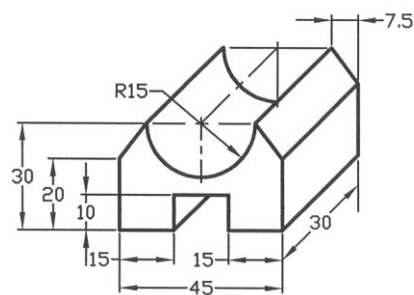
Solution of P6.44



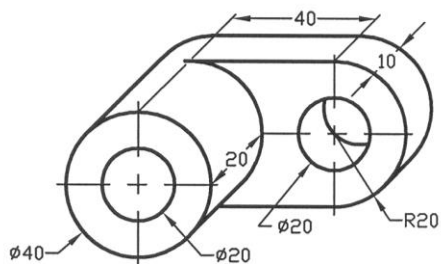
Solution of P6.45



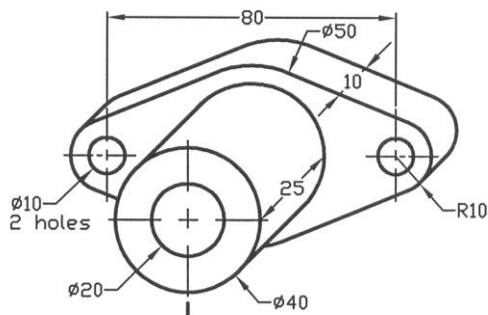
Solution of P6.46



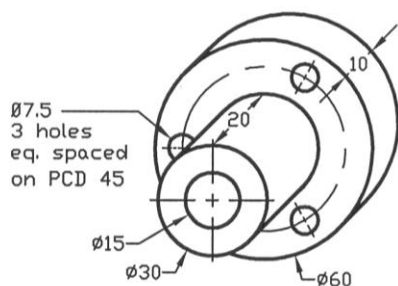
Solution of P6.47



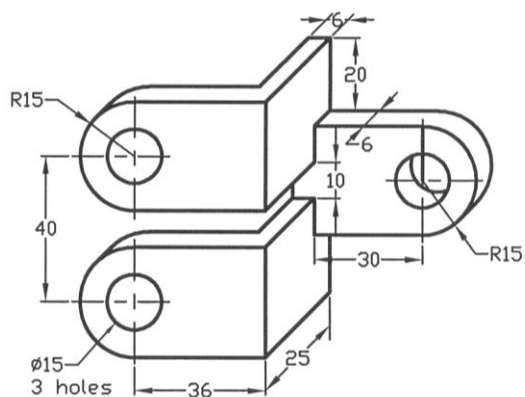
Solution of P6.48



Solution of P6.49



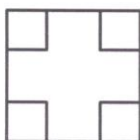
Solution of P6.50



Solution of P6.51

Problems

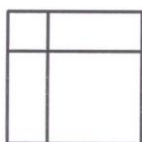
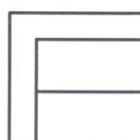
Prob. 6.52- 6.67: Draw the free hand isometric sketches of the objects from the following views as shown in Fig. P6.52 to P6.67.



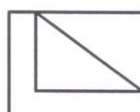
Prob. P6.52



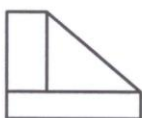
Prob. P6.53

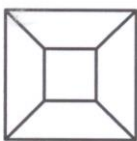


Prob. P6.54

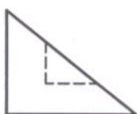
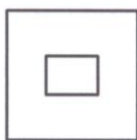


Prob. P6.55

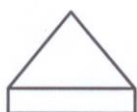
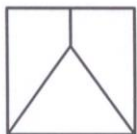




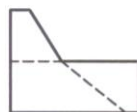
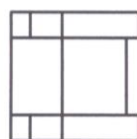
Prob. P6.56



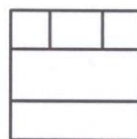
Prob. P6.58



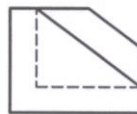
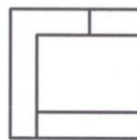
Prob. P6.60



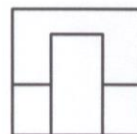
Prob. P6.62



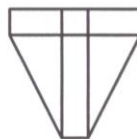
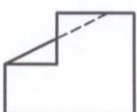
Prob. P6.57



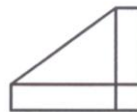
Prob. P6.59

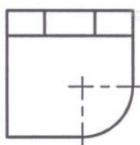
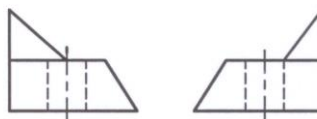
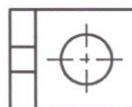


Prob. P6.61

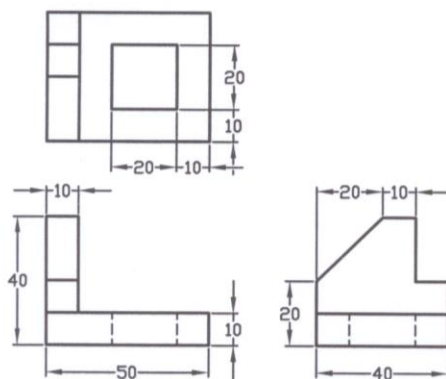
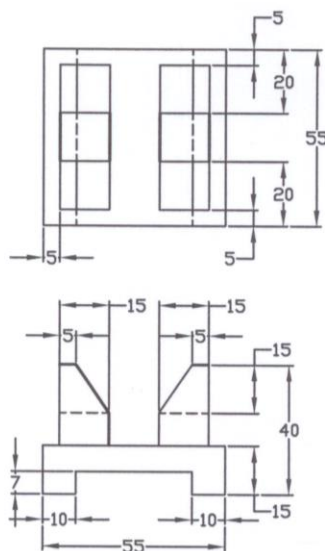


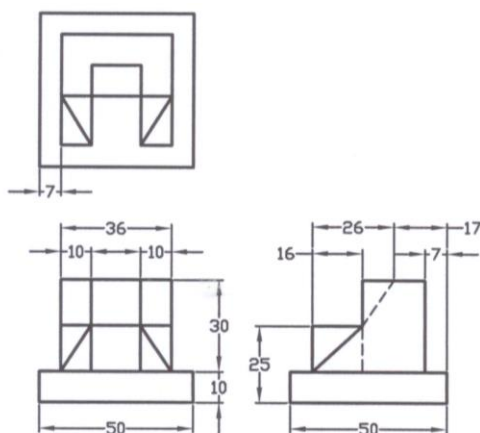
Prob. P6.63



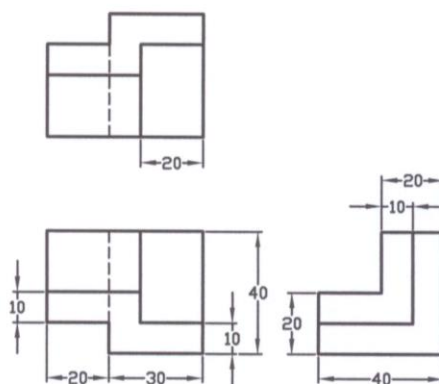
**Prob. P6.64****Prob. P6.65****Prob. P6.66****Prob. P6.67**

Prob. 6.68 - 6.85: Draw the isometric drawing of the objects from the following views as shown in Fig. P6.68 to P6.85.

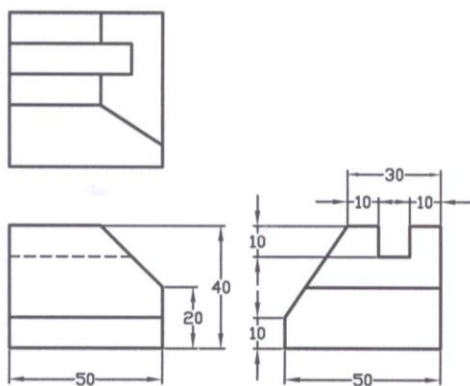
**Prob. P6.68****Prob. P6.69**



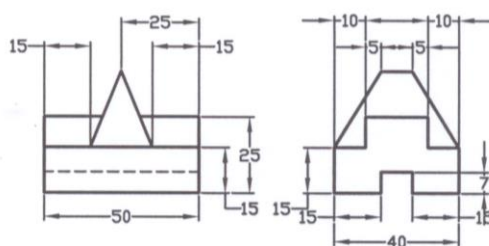
Prob. P6.70



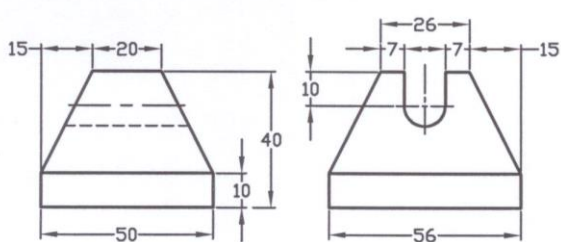
Prob. P6.71



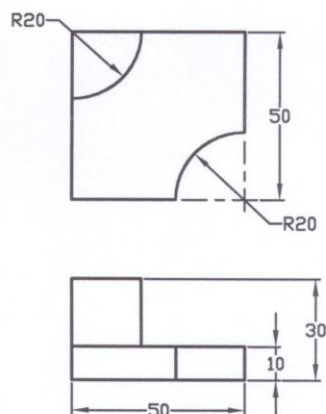
Prob. P6.72



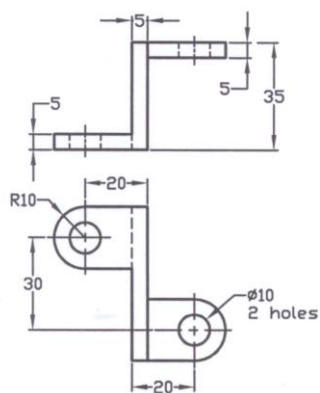
Prob. P6.73



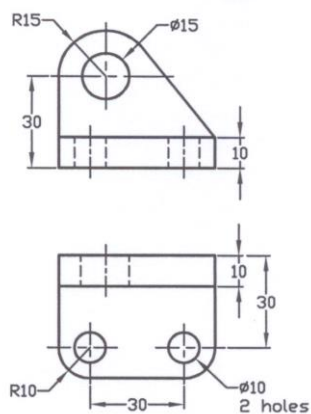
Prob. P6.74



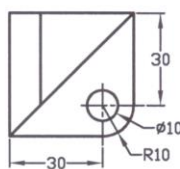
Prob. P6.75



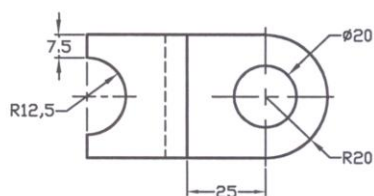
Prob. P6.76



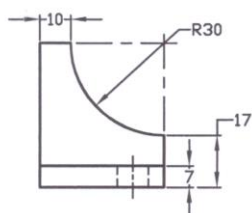
Prob. P6.77



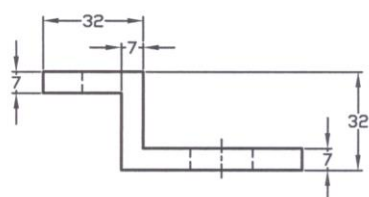
Prob. P6.78



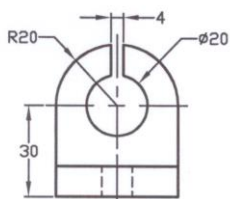
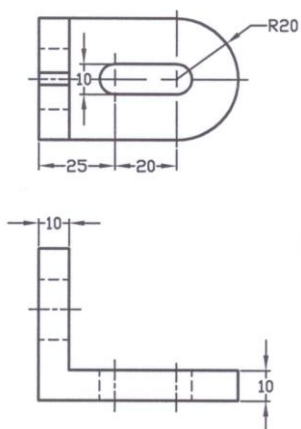
Prob. P6.79

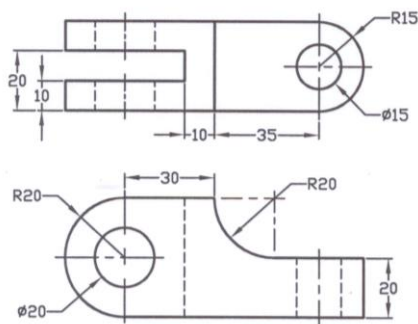
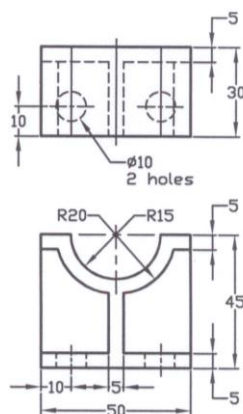
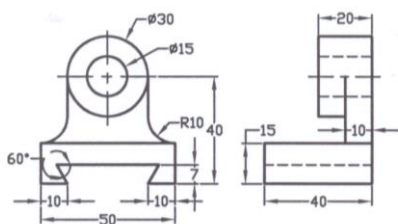
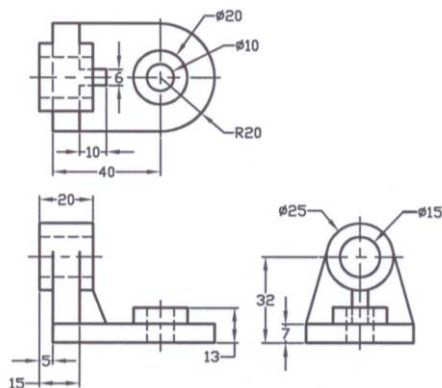


Prob. P6.80

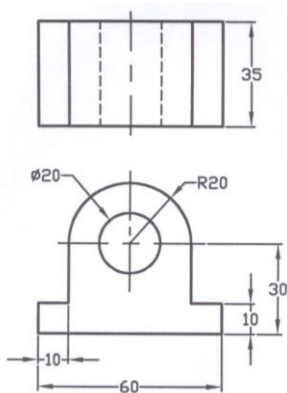
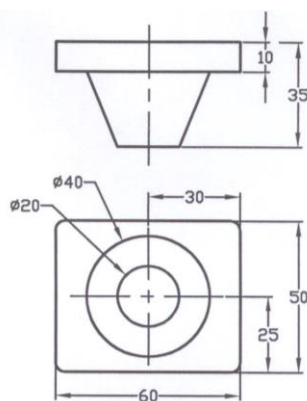


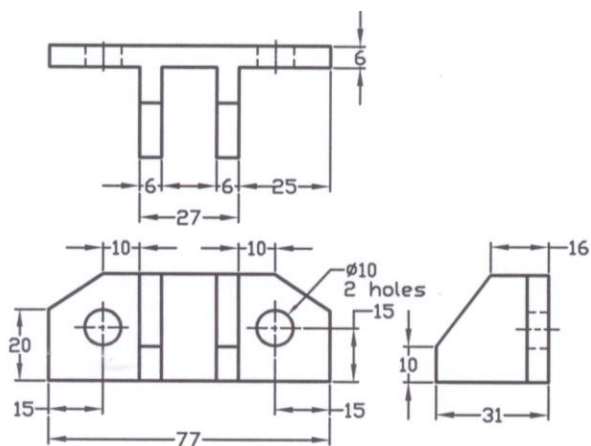
Prob. P6.81



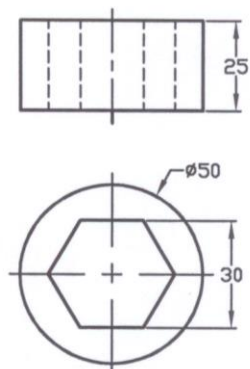
**Prob. P6.82****Prob. P6.83****Prob. P6.84****Prob. P6.85**

Prob. 6.86- 6.91: Draw the oblique views (cavalier) of the objects from the following views as shown in Fig. P6.86 to P6.91.

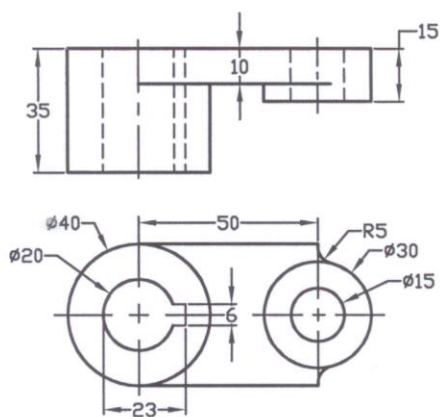
**Prob. P6.86****Prob. P6.87**



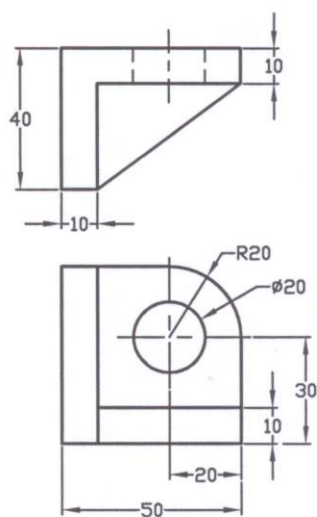
Prob. P6.88



Prob. P6.89



Prob. P6.90



Prob. P6.91